

Cancer Research Support System for Information Scientists: Examples of DNAm Data Analysis and Cell Signaling Simulation

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Background and Aim

It is difficult for information scientists without biology background to start cancer research. The main problem is that cancer is very complicated and it is hard to know “what is cancer?” or “what is the core issue of cancer research”.

→ This study provides two examples as a part of support system for information scientists and students. These two examples would be a key to understand the basis of cancer.

→ More information scientists could work on cancer and more information technologies could be used for cancer research.

System Outline

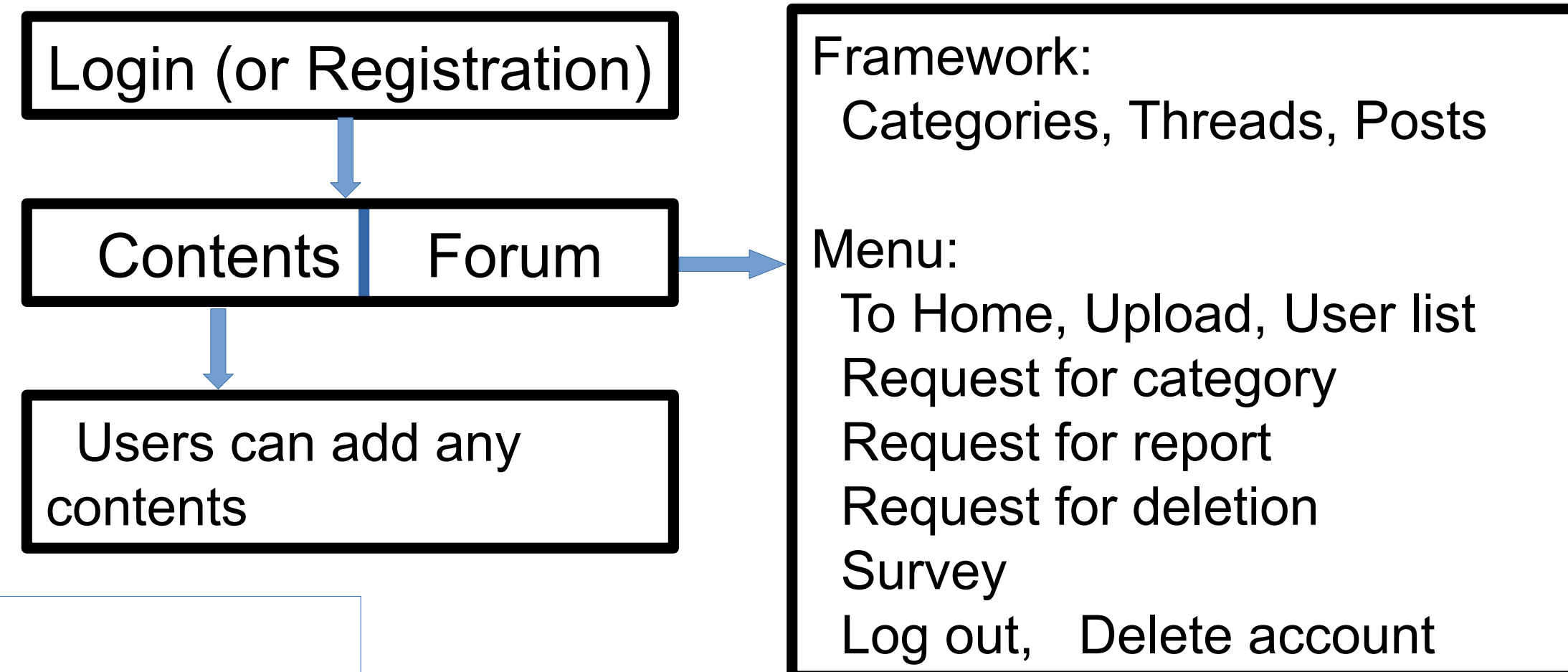
This support system consists of a forum and contents. The forum provides several communication tools for discussions. The contents provide research examples.

(a) DNA Methylation Data Analysis Example: Analysis has two parts.

- Selecting important genes to classify tumor/non-tumor using Decision Tree.
- Evaluating these genes using Neural Network.

(b) Cell Signaling Simulation Example: Modeling the concentration of active RAS using ordinary differential equations.

Overview of the system



DNA Methylation Data Analysis Example

[Methods]

Data and Feature Vector We used two TCGA-STAD datasets (Data1 and Data2).

Data1: Dataset ID: TCGA-STAD.methylation27.tsv, Platform: Illumina Human Methylation 27

Samples: 73 (tumor:48, nontumor:25), Version: 05-09-2024, Type of data: DNA methylation, Unit: beta value

Data2: Dataset ID: TCGA.STAD.sampleMap/HumanMethylation27, Platform: Illumina Infinium HumanMethylation27

Samples: 141 (tumor:81, nontumor:59), Version: 2017-09-08, Type of data: DNA methylation, Unit: beta value

Feature Vector of Decision Tree: Sequence of beta values of genes (of Panel1, Panel2) or all CpG sites of illumina27k using dataset of Data1 or Data2. e.g. “Panel1 X Data1” means Data1 is restricted to genes of Panel1.

For Panel1/Panel2 in Data1, the mean beta value is computed for all CpG sites associated with the same gene.

Panel1: QIAseq Targeted DNA Panel's 63 genes [1]

Panel2: Guardant360's 74 genes

Analysis

Development language&library: Python, scikit-learn

Step1: Decision Tree

Input: Feature vector of beta values of each data (e.g. Panel1 X Data1).

Output: Important genes to classify tumor/non-tumor (average of 900 classifications).

Step 2: Neural Network

Input: Feature vector of beta values of important genes.

Output: Classification of tumor/non-tumor and the accuracy (average of 200 classifications)

[Results]

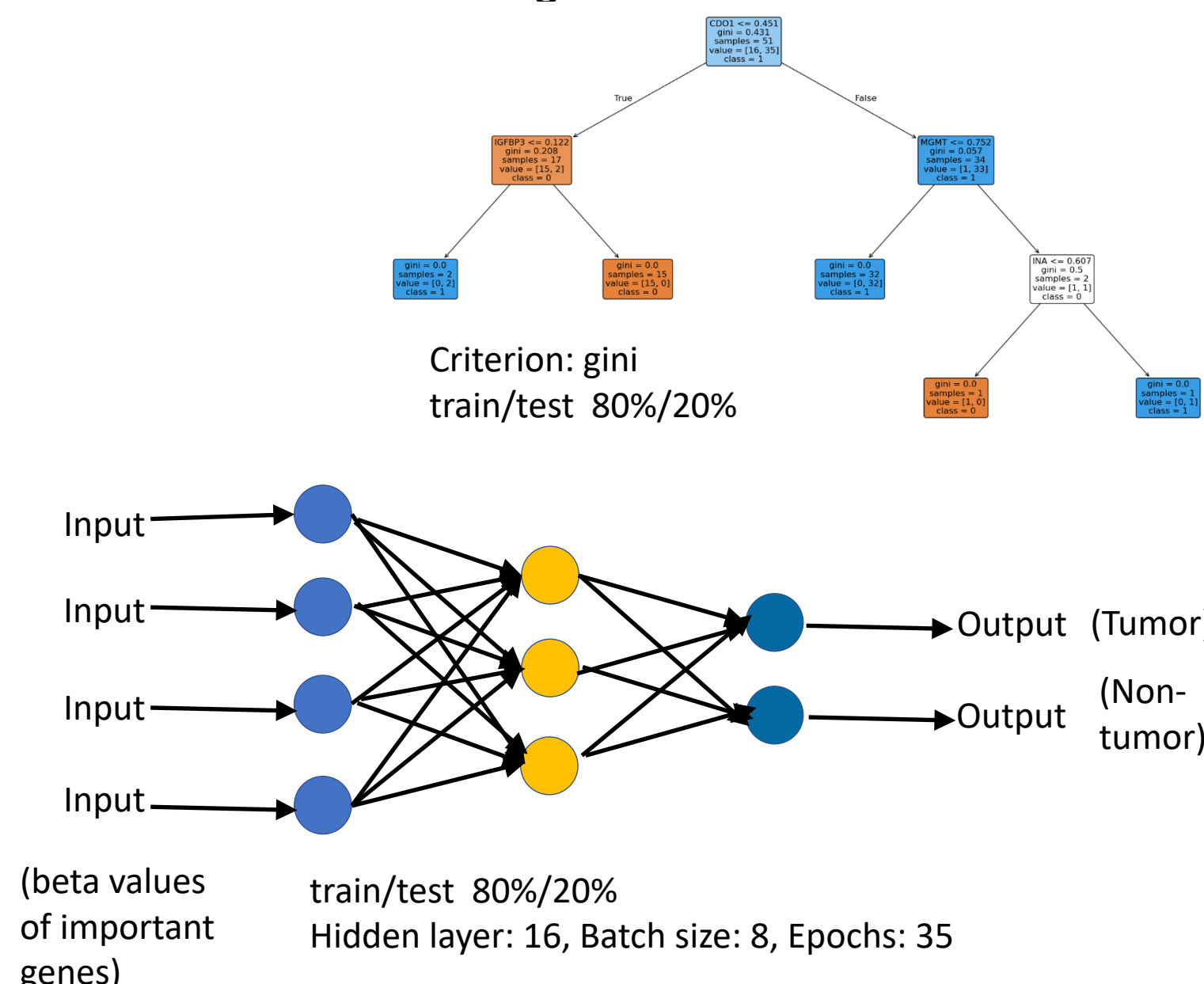
Important genes by using Decision Tree:

Table 1.1 Important genes of this research by using Decision Tree

Data	Important genes
Panel1 X Data1	CDO1, INA, SHOX2, RUNX3, MGMT, IRF4, SOCS1, BMP3
Panel1 X Data2	CDO1, INA, PDGFRA, SHOX2, MGMT, IRF4, THBS1, RUNX3
Panel2 X Data1	CDH1, FGFR2, DDR2, SMAD4, ERBB2, APC, EZH2, CDKN2A
Panel2 X Data2	CDH1, ERBB2, ESR1, PTEN, APC, BRCA1, BRCA2, HRAS
All CpG sites X Data1	TNK1, LAD1, EVI, FOXQ1, CST3, CDO1, OPRM1, BCL2

Table 1.2 Important genes of preceding research

Research	Important genes
O Univ. [1]	SPG20, FBN1, SDC2, TFPI2, SEPT9
I Univ. [2]	GYPC, MME, PRICKLE2, ANGPT1, BMP3, CPNE5, IRF4, KCNB2



[Discussion of the Analysis]

- Data of this research is small (this research did not use the data of illumina450k).
- Table 1.1 shows that the important genes are different among data sets (and restrictions).
- The important genes of this research are not evaluated by real patients' data, but they are compared with the results of preceding research.
- The important genes common to I Univ. and this research are IRF4 and BMP3. There is no important gene common to O Univ. and I Univ. / this research.
- The common genes of Panel1 and Panel2 are APC, CDKN2A, MLH1 and PDGFRA. APC, CDKN2A and PDGFRA are important genes of this research.
- Three important genes of this research “RUNX3 from Panel1, CDH1 and CDKN2A from Panel2” are known as genes related to gastric cancer, while these genes are not included in the results of O Univ. / I Univ.
- Table 1.3 shows the results of All CpG sites are slightly higher than others.
- The classification accuracies are almost the same among several sets of important genes. This result suggests the difficulty of liquid biopsy.
- Users can study the basics of DNA methylation data analysis by tracing the process of this example.

Conclusion

- This research provides forum system and two examples for information scientists without biology background and students.
- One example is on DNA methylation data analysis: Selecting important genes using Decision Tree and Evaluating these genes using Neural Network.
- The results shows that this method can find several important genes.
- Another example is on Cell signaling simulation: Modeling the concentration using ordinary differential equations.
- Users can study some knowledge about DNA methylation data analysis and the basics of Signal transduction.

Acknowledgment

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Cell Signaling Simulation Example

[Methods]

Signaling Pathway: MAPK

Modeling: The concentration of active Ras is modeled by ordinary differential equations.

Simulator:

-Development languages&libraries: HTML, CSS, JavaScript, Python (Pyodide), NumPy, SciPy (LSODA)

-Usage: Users can select a model, set several parameters and simulate the model.

Explanations: The basics of Signal transduction, Activation of EGFR, and Mathematical model are explained.

[Results]

Stimulus : Steps in the concentration of active EGFR (x_{tot}) (Fig. 2.1).

Parameters: $k_1=k_2=k_3=1$; $k_{-1}=k_{-2}=0.1$; $k_{-3}=10$; $y_{tot}=1$.

Resulting Dynamics : Pulses of concentration of active Ras (y^*) (Fig. 2.2).

The system returns to the same baseline level.

This result is the same as described in reference [3].

$$\frac{dx_2}{dt} = k_1(x_{tot} - x_2 - x_3) - k_{-1}x_2 - k_2x_2 + k_{-2}x_3$$

$$\frac{dx_3}{dt} = k_2x_2 - k_{-2}x_3$$

$$\frac{dy^*}{dt} = k_3x_2(y_{tot} - y^*) - k_{-3}x_3 \cdot y^*$$

[Discussion of the Simulation and Forum]

- Users can study the basics of Signal transduction.
- Users can use the simulator to understand the Math model.
- Users can modify the program for building another model.
- Users can build a Forum for their research community and can share their contents.

References

- [1] Nagano, S et al.: Sci Rep 14, 30739, 2024. DOI: 10.1038/s41598-024-79252-y
- [2] Hosseini M et al.: J Cell Mol Med, 27(5), 2023. DOI: 10.1111/jcmm.17693
- [3] J. Ferrell: ISBN: 9780815346036, 2021.

Future work

New application of information technology to cancer research:

- Multi-agent system to solve the problem of Drug Resistance.
- Knowledge representation of Metastasis (TME, EVs, etc.)

This system will be released on GitHub or on our Website:

- Forum (without contents)
- Example (DNAm Data Analysis)
- Example (Cell Signaling Simulation)

If you have any comments on this research or future work, please fill in the form:

<https://forms.gle/CrC35zDjxnnwSsuz7>



COI Disclosure Information

Lead Presenter/Principal

Researcher: Sachiko Deguchi

I have no financial relationships to disclose.

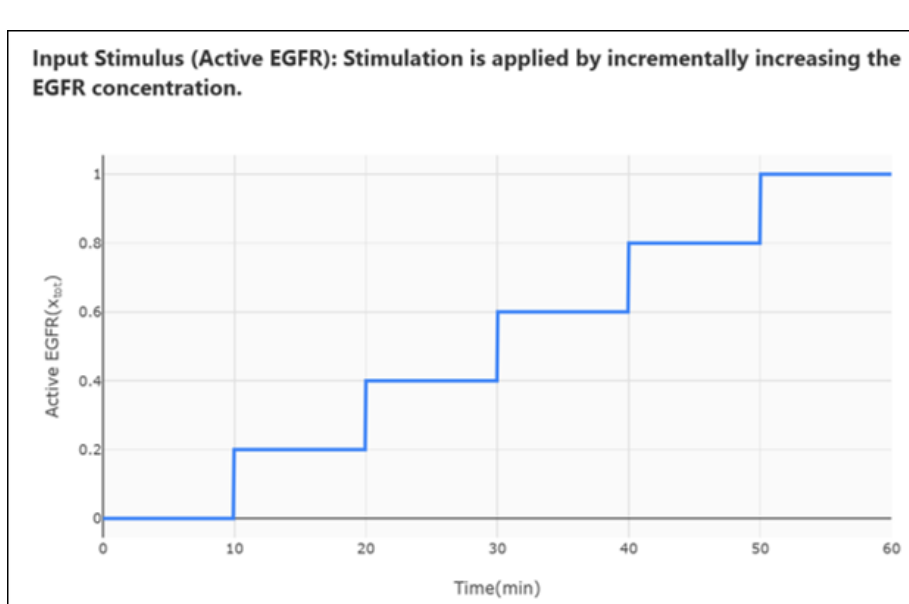
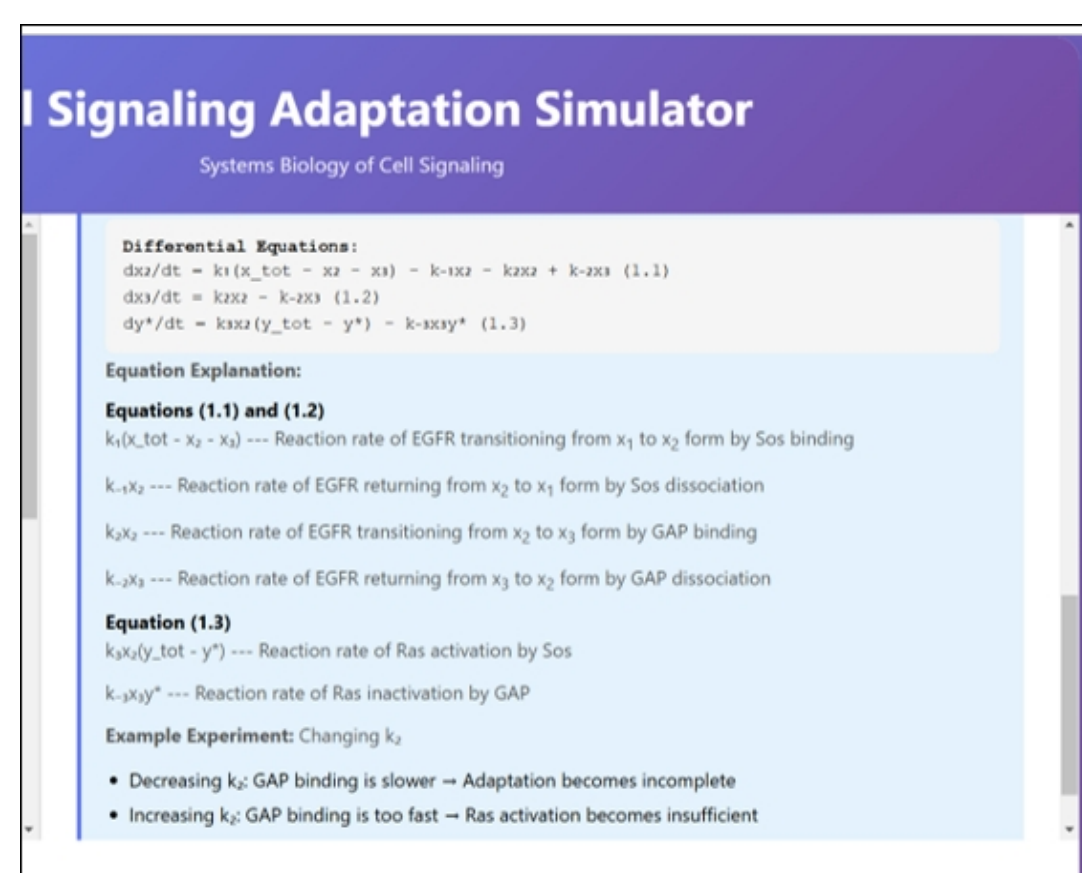
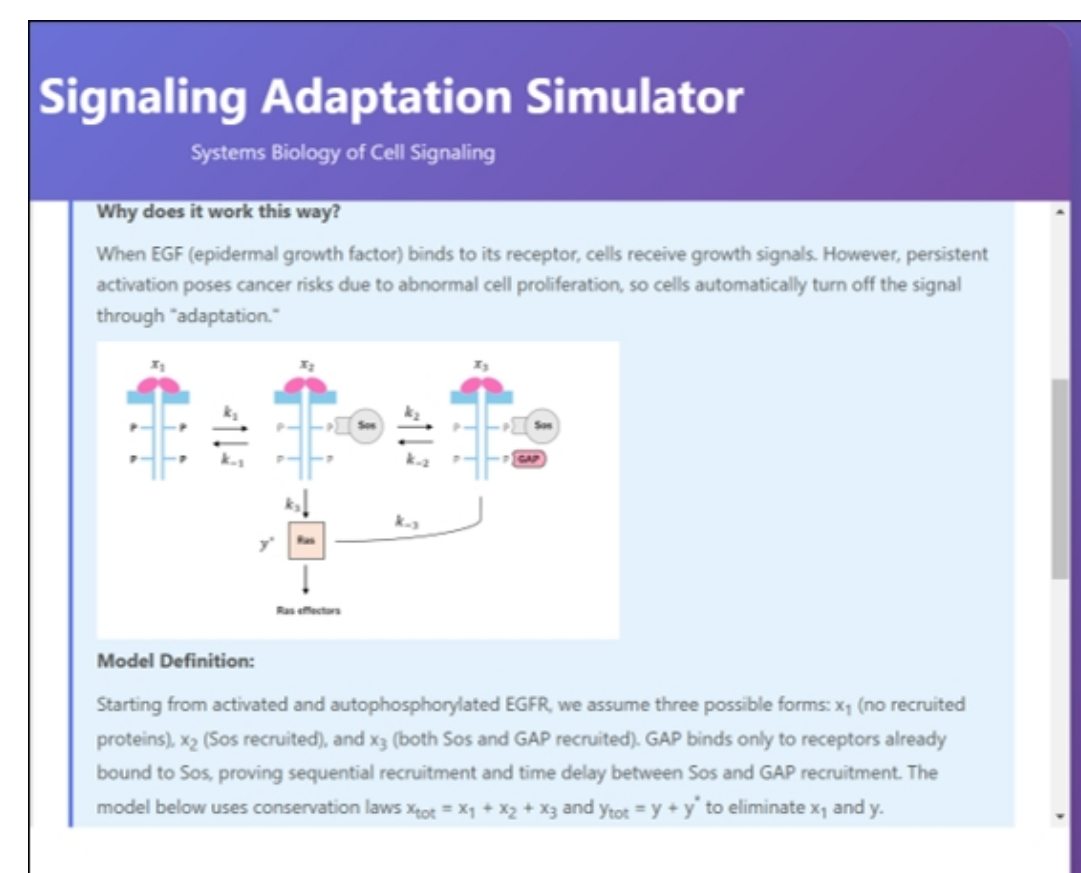


Fig. 2.1 active EGFR

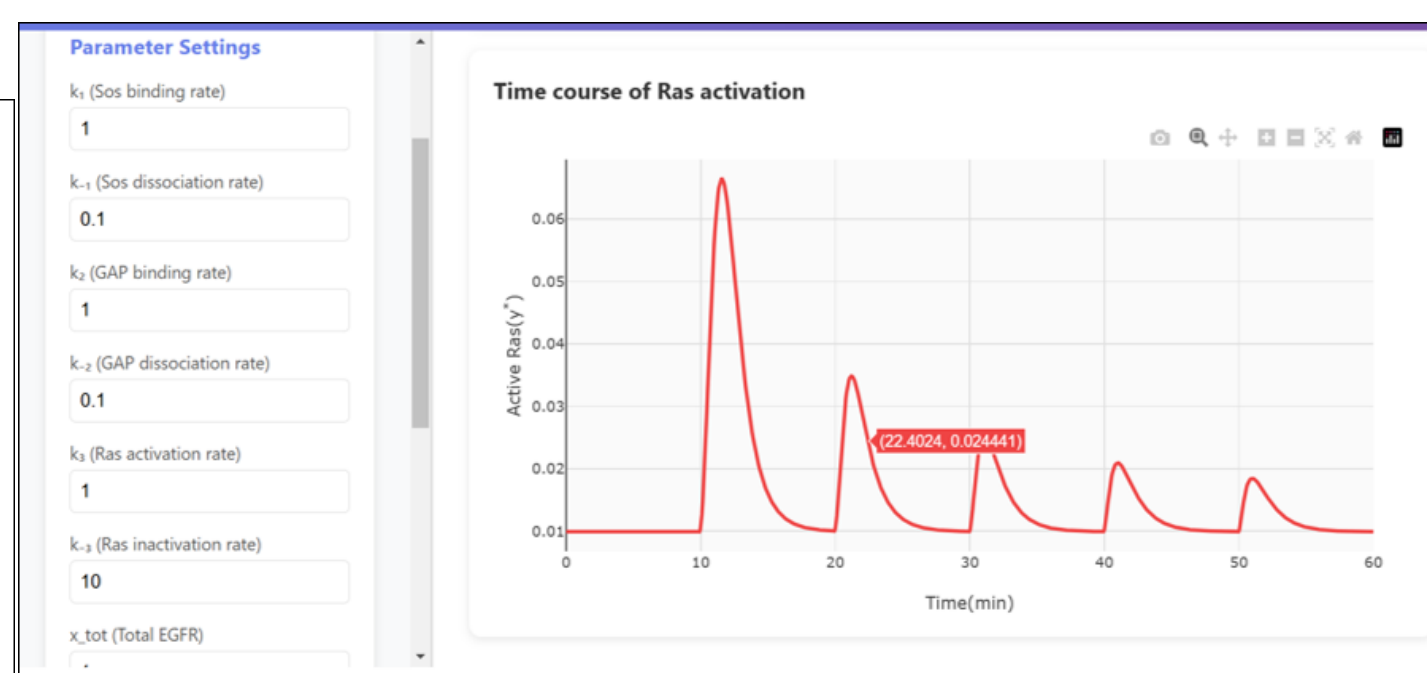


Fig. 2.2 active Ras